

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

“Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training”

Lab Title: SBT-DF203 Lab 7: DNS Introduction and Traffic Analysis

Course Name: Basic Networking Skills for Digital Forensics

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Operating Platform: Kali Linux 2026 (Rolling Release)

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Configured Resolver: 192.168.199.2 (Port 53 UDP)

Report Version: Version 1.0 (Final Release)

Executive Summary

This report documents a normal DNS forensic baseline. The configured resolver (192.168.199.2) was identified. Controlled dig queries for A, AAAA, MX, and NS records were executed and analyzed. A fresh DNS capture was generated and SHA-256 hashed, and packet-level query/response fields were extracted with matching transaction IDs. Browser-generated DNS traffic was inventoried, and four DNS answer IPs were correlated with subsequent TCP connections (ports 443 and 80). An optional SMTP DNS correlation from Lab 4 confirmed mail.patriots.in → 74.53.140.153, matching the SMTP server previously analyzed.

1. Resolver and Environment

- Host Platform:** Kali Linux 2026 Virtual Machine.
- Active Network Interface:** eth0 (IPv4: 192.168.199.128/24, MAC: 00:0c:29:f3:f7:44).
- Configured Name Resolver:** 192.168.199.2 (Identified via /etc/resolv.conf).
- Default Gateway:** 192.168.199.2.

2. Evidence Integrity

Cryptographic verification hashes computed for the primary DNS captures:

Evidence Capture File	SHA-256 Hash Digest	Integrity Status
fresh_dig_dns.pcapng	1a1a09c8ca8f0e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be	VERIFIED
browser_dns.pcapng	fbf9dbb7a619d1c8024971ea6178f82219f27875d57c019cfdb88f3c9991cec0	VERIFIED

3. dig Results

- A Record Query (example.com):** Status: NOERROR, TX ID: 45906, Query time: 39ms. Answers: 104.20.23.154, 172.66.147.243 (TTL: 5 seconds).
- AAAA Record Query (example.com):** Status: NOERROR, TX ID: 19826, Query time: 31ms. Answers: 2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a (TTL: 5 seconds).

- **MX Record Query (example.com):** Status: NOERROR, TX ID: 48349, Query time: 2032ms. Answer: MX 0 . (Null MX record indicating domain does not accept mail).
- **NS Record Query (example.com):** Status: NOERROR, TX ID: 59002, Query time: 2032ms. Answers: elliott.ns.cloudflare.com, hera.ns.cloudflare.com (TTL: 5 seconds).

Protocol Definitions: Core DNS Resource Record Types:

- **A (Address):** Maps a fully qualified domain name (FQDN) to a 32-bit IPv4 address.
- **AAAA (IPv6 Address):** Maps an FQDN to a 128-bit IPv6 address.
- **MX (Mail Exchange):** Designates host responsible for receiving email on behalf of the domain, paired with priority weighting.
- **NS (Name Server):** Delegates a DNS zone to authoritative name servers.
- **TTL (Time-to-Live):** Specifies the duration (in seconds) a resolver or client may cache the record before issuing a new resolution. It is a cache expiry timer, NOT a record creation timestamp.

4. Packet-Level Query/Response Analysis

Dissection of fresh_dig_dns.pcapng proves exact parity between outbound client queries and incoming resolver responses:

Frame	Client Port	Query Name	Type	Transaction ID
1	55170	example.com	A (1)	0x1865
3	47703	example.com	AAAA (28)	0xd754
5	36831	google.com	A (1)	0x895d

Frame	TX ID	RCODE	Returned Answers	TTL
2	0x1865	NOERROR	172.66.147.243, 104.20.23.154	5
4	0xd754	NOERROR	2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a	5
6	0x895d	NOERROR	142.251.142.14	5

- **Matching Transaction IDs:** Every outbound query's transaction ID (e.g., 0x1865) matched the inbound response ID exactly, proving request-response pair correlation across UDP port 53.

5. Browser DNS Inventory

A standard browser visit to http://example.com generated 33 unique DNS query names across telemetry, fonts, certificate validation, and CDN domains:

- **Sample Discovered Domains:** example.org, www.google.com, detectportal.firefox.com, contile.services.mozilla.com, push.services.mozilla.com, firefox.settings.services.mozilla.com, fonts.gstatic.com, csp.withgoogle.com, play.google.com.
- **Forensic Finding:** A single user-initiated webpage access triggers an extensive cascade of background lookups for content delivery networks, static assets, telemetry, and security checks.

6. Connection Correlation (Part F & Part G)

Correlating DNS A-record answers against subsequent outbound TCP connection attempts (SYN packets with ACK=0):

DNS Answer IP	Observed TCP Destination	Correlation Verdict
140.248.169.91	140.248.169.91:443 / 80	EXACT MATCH
172.66.147.243	172.66.147.243:443	EXACT MATCH
34.107.243.93	34.107.243.93:443	EXACT MATCH
34.149.226.178	34.149.226.178:443	EXACT MATCH

- **Definitive Correlation Proof:** All 4 DNS answer IPs appeared in subsequent TCP SYN packets, providing conclusive evidence that the host resolved those domains and established communication.
- **SMTP DNS Cross-Correlation (Part G):** Analysis of preserved Lab 4 evidence revealed: Query mail.patriots.in -> Answer 74.53.140.153 (TTL 1082), matching the Exim SMTP mail server investigated in Lab 4.

7. Normal Variations Observed

- **Multiple A/AAAA Answers:** Queries returned multiple IP addresses per request (e.g., Cloudflare Anycast CDN load-balancing).
- **Ultra-Low TTL (5 seconds):** CDN-managed domains utilize short TTLs to enable rapid traffic rerouting and failover.
- **IPv6 Lookups Without IPv6 Connections:** The browser automatically requested AAAA records, but connecting sockets were exclusively IPv4.
- **Browser Telemetry and Probes:** Observed captive portal checks (detectportal.firefox.com) and IPv4-only ARPA queries.

8. Limitations

- **Upstream Capture URL 404:** The external dig_dns.pcap URL returned HTTP 404; a locally generated capture was used instead, providing repeatable evidence.
- **IPv6 Connectivity:** The virtual laboratory lacked native IPv6 routing, restricting socket connections to IPv4.
- **Cross-Lab Evidence Retention:** The SMTP correlation was made possible by properly maintaining chain of custody on Lab 4 captures.

9. Screenshots and Evidence

The following 15 figures document the complete DNS baseline investigation, dig queries, packet captures, browser inventories, and TCP connection correlation:

```
(kali@kali)-[~/SBT-DF203-Lab6]
└─$ cd ~
└─$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
└─$ cd ~/SBT-DF203-Lab7

(kali@kali)-[~/SBT-DF203-Lab7]
└─$ pwd
/home/kali/SBT-DF203-Lab7
└─$ find . -maxdepth 1 -type d -print
./
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)-[~/SBT-DF203-Lab7]
└─$ sudo apt update
└─$ sudo apt install -y dnsutils tshark wireshark
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 77.0 MB in 24min 34s (52.2 kB/s)
2092 packages can be upgraded. Run 'apt list --upgradable' to see them.
[sudo] password for kali:
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0  libcrypt-dev  libkmlbase1t64  libmplex2-2.1-0t64  libqt6test6  python3-fs  rpcsvc-proto
  libann0  libgavi-1  libkmlcore1t64  libpoppler147  libqt6xml6  python3-gpg  samba-ad-dc
  libarmadillo14  libgdal37  libkmlengine1t64  libportmidi0  libraw23t64  python3-pyqt6.sip  samba-ad-provision
  libavc1394-0  libgirepository-1.0-1  liblab-gamut1  libpython3.13-dev  libwireshark18  python3.13  samba-dsdb-modules
```

Figure 1: Directory Setup & Package Installations — Creating /SBT-DF203-Lab7 hierarchy and installing bind9-dnsutils, TShark, and Wireshark.

```
(kali@kali)-[~/SBT-DF203-Lab7]
└─$ cd ~/SBT-DF203-Lab7
└─$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvconf status 2>/dev/null | tee reports/resolvconf_status.txt || true
systemd-resolve --status 2>/dev/null | tee reports/systemd_resolve_status.txt || true
└─$ cat reports/resolv_conf.txt
# Generated by NetworkManager
search localdomain
nameserver 192.168.199.2
# Generated by NetworkManager
search localdomain
nameserver 192.168.199.2

(kali@kali)-[~/SBT-DF203-Lab7]
└─$ cd ~/SBT-DF203-Lab7
└─$ wget -O evidence/dig_dns.pcap 'https://raw.githubusercontent.com/frankxwu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap' 2>&1 |
└─$ tee reports/download_attempt.txt
ls -lh evidence/dig_dns.pcap 2>/dev/null
--2026-09-21 17:30:17-- https://raw.githubusercontent.com/frankxwu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-21 17:30:19 ERROR 404: Not Found.

-rw-r--r-- 1 kali kali 0 Sep 21 17:30 evidence/dig_dns.pcap

(kali@kali)-[~/SBT-DF203-Lab7]
└─$ ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
└─$ cat reports/interfaces.txt
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.199.128/24 fe80::8c84:5c9c:8758:3178/64
default via 192.168.199.2 dev eth0 proto dhcp src 192.168.199.128 metric 100
192.168.199.0/24 dev eth0 proto kernel scope link src 192.168.199.128 metric 100
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.199.128/24 fe80::8c84:5c9c:8758:3178/64

(kali@kali)-[~/SBT-DF203-Lab7]
└─$
```

Figure 2: Resolver Configuration & Interface Routing Inspection — Inspecting /etc/resolv.conf (192.168.199.2), 404 capture note, interface eth0, and routing tables.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
dig example.com A | tee reports/dig_example_A.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 45906
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                5       IN      A      104.20.23.154
example.com.                5       IN      A      172.66.147.243

;; Query time: 39 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:36:26 EDT 2026
;; MSG SIZE rcvd: 72

(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 19826
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:

```

Figure 3: Controlled dig Queries: A and AAAA Records — Executing dig example.com A (104.20.23.154, 172.66.147.243) and AAAA records.

```

;; ANSWER SECTION:
example.com.          5      IN      AAAA    2606:4700:10::ac42:93f3
example.com.          5      IN      AAAA    2606:4700:10::6814:179a

;; Query time: 31 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:36:38 EDT 2026
;; MSG SIZE rcvd: 96

(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com MX | tee reports/dig_example_MX.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 48349
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.          5      IN      MX      0 .

;; Query time: 2032 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:36:51 EDT 2026
;; MSG SIZE rcvd: 55

(kali@kali)-[~/SBT-DF203-Lab7]
$ dig example.com NS | tee reports/dig_example_NS.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 59002

```

Figure 4: *Controlled dig Queries: MX and NS Records* — Executing dig example.com MX (Null MX) and dig example.com NS (Cloudflare name servers).

```

;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 59002
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                5       IN      NS      elliottns.cloudflare.com.
example.com.                5       IN      NS      herans.cloudflare.com.

;; Query time: 2032 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:37:03 EDT 2026
;; MSG SIZE rcvd: 95

(kali@kali)-[~/SBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
cat reports/dig_example_short.txt
172.66.147.243
104.20.23.154
172.66.147.243
104.20.23.154

(kali@kali)-[~/SBT-DF203-Lab7]
$ echo "=== SERVER (Resolver) ===" | tee reports/dig_summary.txt
grep -E '^;; SERVER' reports/dig_example_A.txt | tee -a reports/dig_summary.txt
echo "" | tee -a reports/dig_summary.txt
echo "=== STATUS/FLAGS ===" | tee -a reports/dig_summary.txt
grep -E '^;; -->HEADER' reports/dig_example_A.txt | tee -a reports/dig_summary.txt
echo "" | tee -a reports/dig_summary.txt
echo "=== QUERY TIME ===" | tee -a reports/dig_summary.txt
grep -E '^;; Query time' reports/dig_example_A.txt | tee -a reports/dig_summary.txt
echo "" | tee -a reports/dig_summary.txt
echo "=== ANSWERS (A) ===" | tee -a reports/dig_summary.txt
grep -A5 'ANSWER SECTION' reports/dig_example_A.txt | tee -a reports/dig_summary.txt
echo "" | tee -a reports/dig_summary.txt
echo "=== ANSWERS (MX) ===" | tee -a reports/dig_summary.txt

```

Figure 5: Short Record Verification & Summary Script Start — Verifying dig +short output and drafting automated diagnostic extraction script.

```

echo "" | tee -a reports/dig_summary.txt
echo "=== ANSWERS (MX) ===" | tee -a reports/dig_summary.txt
grep -A5 'ANSWER SECTION' reports/dig_example_MX.txt | tee -a reports/dig_summary.txt
echo "" | tee -a reports/dig_summary.txt
echo "=== ANSWERS (NS) ===" | tee -a reports/dig_summary.txt
grep -A8 'ANSWER SECTION' reports/dig_example_NS.txt | tee -a reports/dig_summary.txt
cat reports/dig_summary.txt
=== SERVER (Resolver) ===
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)

=== STATUS/FLAGS ===
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 45906

=== QUERY TIME ===
;; Query time: 39 msec

=== ANSWERS (A) ===
;; ANSWER SECTION:
example.com.      5      IN      A      104.20.23.154
example.com.      5      IN      A      172.66.147.243

;; Query time: 39 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)

=== ANSWERS (MX) ===
;; ANSWER SECTION:
example.com.      5      IN      MX      0 .

;; Query time: 2032 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:36:51 EDT 2026

=== ANSWERS (NS) ===
;; ANSWER SECTION:
example.com.      5      IN      NS      elliottns.cloudflare.com.
example.com.      5      IN      NS      herans.cloudflare.com.

;; Query time: 2032 msec
;; SERVER: 192.168.199.2#53(192.168.199.2) (UDP)
;; WHEN: Mon Sep 21 17:37:03 EDT 2026
;; MSG SIZE rcvd: 95

```

Figure 6: *dig Diagnostic Summary Report* — Compiling resolver IP, status codes (NOERROR), query execution times, and answers.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
IFACE=eth0
tshark -i $IFACE -f 'port 53' -w evidence/fresh_dig_dns.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
6 ^C

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
cp --preserve=timestamp evidence/fresh_dig_dns.pcapng working/fresh_dig_dns_working.pcapng
sha256sum evidence/fresh_dig_dns.pcapng working/fresh_dig_dns_working.pcapng | tee reports/fresh_dns_sha256.txt
cat reports/fresh_dns_sha256.txt
1a1a09c8ca8f8e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be evidence/fresh_dig_dns.pcapng
1a1a09c8ca8f8e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be working/fresh_dig_dns_working.pcapng
1a1a09c8ca8f8e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be evidence/fresh_dig_dns.pcapng
1a1a09c8ca8f8e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be working/fresh_dig_dns_working.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
PCAP=working/fresh_dig_dns_working.pcapng
tshark -r "$PCAP" -Y 'dns.flags.response == 0' -T fields -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport -e dns.id -e dns.qr
y.name -e dns.qry.type | tee reports/dns_queries.tsv
cat reports/dns_queries.tsv
Warning: program compiled against libxml 215 using older 214
1 2026-09-21T17:43:07.053802973-0400 192.168.199.128 55170 192.168.199.2 53 0x1865 example.com 1
3 2026-09-21T17:43:07.130276185-0400 192.168.199.128 47703 192.168.199.2 53 0xd754 example.com 28
5 2026-09-21T17:43:09.105283786-0400 192.168.199.128 36831 192.168.199.2 53 0x895d google.com 1
1 2026-09-21T17:43:07.053802973-0400 192.168.199.128 55170 192.168.199.2 53 0x1865 example.com 1
3 2026-09-21T17:43:07.130276185-0400 192.168.199.128 47703 192.168.199.2 53 0xd754 example.com 28
5 2026-09-21T17:43:09.195283786-0400 192.168.199.128 36831 192.168.199.2 53 0x895d google.com 1

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r "$PCAP" -Y 'dns.flags.response == 1' -T fields -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport -e dns.id -e dn
s.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl | tee reports/dns_responses.tsv
cat reports/dns_responses.tsv
Warning: program compiled against libxml 215 using older 214
2 2026-09-21T17:43:07.089287833-0400 192.168.199.2 53 192.168.199.128 55170 0x1865 0 2 172.66.147.243,104.20.23.154 5
,5
4 2026-09-21T17:43:09.166353051-0400 192.168.199.2 53 192.168.199.128 47703 0xd754 0 2 2606:4700:10::ac42:93f3,2606
:4700:10::6814:179a 5,5
6 2026-09-21T17:43:09.219791933-0400 192.168.199.2 53 192.168.199.128 36831 0x895d 0 1 142.251.142.14 5
2 2026-09-21T17:43:07.089287833-0400 192.168.199.2 53 192.168.199.128 55170 0x1865 0 2 172.66.147.243,104.20.23.154 5
,5
4 2026-09-21T17:43:09.166353051-0400 192.168.199.2 53 192.168.199.128 47703 0xd754 0 2 2606:4700:10::ac42:93f3,2606
:4700:10::6814:179a 5,5
6 2026-09-21T17:43:09.219791933-0400 192.168.199.2 53 192.168.199.128 36831 0x895d 0 1 142.251.142.14 5

```

Figure 7: Fresh DNS Capture & SHA-256 Hashing (fresh_dig_dns.pcapng) — Sniffing UDP port 53 traffic, calculating SHA-256 hash, and extracting queries.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r "$PCAP" -Y 'dns.flags.response == 1' -T fields -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport -e dns.id -e dn
s.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl | tee reports/dns_responses.tsv
cat reports/dns_responses.tsv
Warning: program compiled against libxml 215 using older 214
2 2026-09-21T17:43:07.089287833-0400 192.168.199.2 53 192.168.199.128 55170 0x1865 0 2 172.66.147.243,104.20.23.154 5
,5
4 2026-09-21T17:43:09.166353051-0400 192.168.199.2 53 192.168.199.128 47703 0xd754 0 2 2606:4700:10::ac42:93f3,2606
:4700:10::6814:179a 5,5
6 2026-09-21T17:43:09.219791933-0400 192.168.199.2 53 192.168.199.128 36831 0x895d 0 1 142.251.142.14 5
2 2026-09-21T17:43:07.089287833-0400 192.168.199.2 53 192.168.199.128 55170 0x1865 0 2 172.66.147.243,104.20.23.154 5
,5
4 2026-09-21T17:43:09.166353051-0400 192.168.199.2 53 192.168.199.128 47703 0xd754 0 2 2606:4700:10::ac42:93f3,2606
:4700:10::6814:179a 5,5
6 2026-09-21T17:43:09.219791933-0400 192.168.199.2 53 192.168.199.128 36831 0x895d 0 1 142.251.142.14 5

```

Figure 8: DNS Responses & Transaction ID Dissection — Extracting response frames, RCODE NOERROR, returned A/AAAA addresses, and TTL values.

```

(kali@kali)~[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
cat > reports/query_response_match.txt << 'EOF'
=== DNS QUERY-RESPONSE CORRELATION ===

TX ID | Query Name | Type | Client Port | Response Frame | RCODE | Answers | TTL
-----|-----|-----|-----|-----|-----|-----|-----
0x1865 | example.com | A | 55170 | 2 | NOERROR | 172.66.147.243, 104.20.23.154 | 5
0xd754 | example.com | AAAA | 47703 | 4 | NOERROR | 2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a | 5
0x895d | google.com | A | 36831 | 6 | NOERROR | 142.251.142.14 | 5
EOF
cat reports/query_response_match.txt
=== DNS QUERY-RESPONSE CORRELATION ===

TX ID | Query Name | Type | Client Port | Response Frame | RCODE | Answers | TTL
-----|-----|-----|-----|-----|-----|-----|-----
0x1865 | example.com | A | 55170 | 2 | NOERROR | 172.66.147.243, 104.20.23.154 | 5
0xd754 | example.com | AAAA | 47703 | 4 | NOERROR | 2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a | 5
0x895d | google.com | A | 36831 | 6 | NOERROR | 142.251.142.14 | 5

(kali@kali)~[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
IFACE=eth0
tshark -i $IFACE -f 'port 53' -w evidence/browser_dns.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
104 ^C

(kali@kali)~[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type | sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt
cat reports/browser_dns_inventory.txt
Warning: program compiled against libxml 215 using older 214
  2 www.google.com 28
  2 www.google.com 1
  2 example.org 1
  2 detectportal.firefox.com 1
  1 www.gstatic.com 28
  1 www.gstatic.com 1

```

Figure 9: Query-Response Correlation Table Display — Verification table confirming matching transaction IDs (0x1865, 0xd754, 0x895d).

```

1 classify-client.services.mozilla.com      1
1 aus5.mozilla.org [REDACTED] 28
1 aus5.mozilla.org                        1
1 addons.mozilla.org                     28
1 addons.mozilla.org                     1
2 www.google.com 28
2 www.google.com 1
2 example.org 1
2 detectportal.firefox.com                1
1 www.gstatic.com 28
1 www.gstatic.com 1
1 versioncheck-bg.addons.mozilla.org      28
1 versioncheck-bg.addons.mozilla.org      1
1 shavar.services.mozilla.com             28
1 shavar.services.mozilla.com             1
1 services.addons.mozilla.org             28
1 services.addons.mozilla.org             1
1 safebrowsing.googleapis.com            28
1 safebrowsing.googleapis.com            1
1 push.services.mozilla.com              28
1 push.services.mozilla.com              1
1 play.google.com 28
1 play.google.com 1
1 ogads-pa.clients6.google.com            28
1 ogads-pa.clients6.google.com            1
1 normandy.cdn.mozilla.net               28
1 normandy.cdn.mozilla.net               1
1 location.services.mozilla.com           28
1 location.services.mozilla.com           1
1 ipv4only.arpa 28
1 ipv4only.arpa 1
1 fonts.gstatic.com                      28
1 fonts.gstatic.com                      1
1 firefox.settings.services.mozilla.com   28
1 firefox.settings.services.mozilla.com   1
1 firefox-settings-attachments.cdn.mozilla.net 28
1 firefox-settings-attachments.cdn.mozilla.net 1
1 example.org 28
1 detectportal.firefox.com                28
1 csp.withgoogle.com                     28
1 csp.withgoogle.com                     1

```

Figure 10: *Browser DNS Query Name Inventory* — Tabulating 33 unique domain lookup queries generated by a single browser session.

```

(kali@kali) [~/SBT-DF203-Lab7]
$ sha256sum evidence/browser_dns.pcapng | tee reports/browser_dns_sha256.txt
cat reports/browser_dns_sha256.txt
fbf9dbb7a619d1c8024971ea6178f82219f27875d57c019cfdb88f3c9991cec0 evidence/browser_dns.pcapng
fbf9dbb7a619d1c8024971ea6178f82219f27875d57c019cfdb88f3c9991cec0 evidence/browser_dns.pcapng

(kali@kali) [~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a | tee reports/dns_A_answers.tsv
cat reports/dns_A_answers.tsv
Warning: program compiled against libxml 215 using older 214
1790027329.388351373 contile.services.mozilla.com 140.248.169.91
1790027329.394024560 content-signature-2.cdn.mozilla.net 34.149.226.178
1790027329.558928812 safebrowsing.googleapis.com 142.251.36.202
1790027332.367177556 push.services.mozilla.com 34.107.243.93
1790027332.684917750 firefox.settings.services.mozilla.com 140.248.169.91
1790027334.335478363 example.org 104.20.26.136,172.66.157.237
1790027334.335479200 example.org 104.20.26.136,172.66.157.237
1790027334.351516541 ipv4only.arpa 102.0.0.170,192.0.0.171
1790027334.356442310 detectportal.firefox.com 140.248.169.91
1790027334.356442744 detectportal.firefox.com 140.248.169.91
1790027336.787036955 firefox-settings-attachments.cdn.mozilla.net 140.248.169.91
1790027337.446823054 www.google.com 142.251.150.119,142.251.156.119,142.251.157.119,142.251.152.119,142.251.153.119,142.251.151.119,142.251.155.119,142.
251.154.119
1790027338.191750023 www.google.com 142.251.154.119,142.251.150.119,142.251.156.119,142.251.157.119,142.251.152.119,142.251.153.119,142.251.151.119,142.
251.155.119
1790027339.537750912 fonts.gstatic.com 142.251.209.131
1790027339.555328490 www.gstatic.com 172.217.22.131
1790027339.582525219 csp.withgoogle.com 142.251.39.113
1790027340.965009147 shavar.services.mozilla.com 54.200.132.34,34.210.242.234,44.231.188.78
1790027342.939508103 ogads-pa.clients6.google.com 142.251.39.202
1790027343.427413856 play.google.com 172.217.22.78
1790027344.905677903 normandy.cdn.mozilla.net 34.49.51.44
1790027345.720682358 consent.google.com 142.251.39.206
1790027345.881790799 classify-client.services.mozilla.com 140.248.169.91
1790027346.494684981 services.addons.mozilla.org 151.101.1.91,151.101.129.91,151.101.65.91,151.101.193.91
1790027346.558535863 location.services.mozilla.com 140.248.169.91
1790027348.792372072 aus5.mozilla.org 146.75.89.91
1790027348.902053790 versioncheck-bg.addons.mozilla.org 34.160.90.233
1790027349.671062330 addons.mozilla.org 151.101.129.91,151.101.1.91,151.101.193.91,151.101.65.91
1790027329.388351373 contile.services.mozilla.com 140.248.169.91

```

Figure 11: Browser DNS Capture Hashing & A-Record Answer Extraction — Computing SHA-256 hash for browser_dns.pcapng and logging returned IPv4 addresses.

```

(kali@kali) [~/SBT-DF203-Lab7]
$ tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn=1 && tcp.flags.ack=0' -T fields -e frame.time_epoch -e ip.dst -e tcp.dstport | tee reports/sub
sequent_tcp_destinations.tsv
cat reports/subsequent_tcp_destinations.tsv
Warning: program compiled against libxml 215 using older 214

(kali@kali) [~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
if [ -f ~/SBT-DF203-Lab4/working/smtp_working.pcap ]; then
  tshark -r ~/SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields -e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp
.name -e dns.resp.ttl | tee reports/smtp_dns_correlation.tsv
  cat reports/smtp_dns_correlation.tsv
else
  echo "SMTP capture not available - documenting as limitation" | tee reports/smtp_dns_correlation.tsv
fi
Warning: program compiled against libxml 215 using older 214
1 2009-10-05T02:06:07.492060000-0400 mail.patriots.in 1
2 2009-10-05T02:06:07.526085000-0400 mail.patriots.in 1 74.53.140.153 mail.patriots.in,patriots.in,patriots.in,patriots.in 1082
7,10828,82828,82828
1 2009-10-05T02:06:07.492060000-0400 mail.patriots.in 1
2 2009-10-05T02:06:07.526085000-0400 mail.patriots.in 1 74.53.140.153 mail.patriots.in,patriots.in,patriots.in,patriots.in 1082
7,10828,82828,82828

```

Figure 12: Subsequent TCP Destinations & Lab 4 SMTP Correlation — Correlating outgoing TCP SYN packets and cross-referencing Lab 4 mail.patriots.in resolution.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
IFACE=eth0
tshark -i $IFACE -w evidence/browser_full.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
229 ^C

(kali@kali)-[~/SBT-DF203-Lab7]
$ cd ~/SBT-DF203-Lab7
tshark -r evidence/browser_full.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a | tee reports/dns_A_answers_full.tsv
tshark -r evidence/browser_full.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields -e frame.time_epoch -e ip.dst -e tcp.dstport | tee reports/subsequent_tcp_destinations.tsv
cat reports/dns_A_answers_full.tsv | head -10
echo ""
cat reports/subsequent_tcp_destinations.tsv | head -10
Warning: program compiled against libxml 215 using older 214
1790028163.021495330 contile.services.mozilla.com 140.248.169.91
1790028163.021495365 content-signature-2.cdn.mozilla.net 34.149.226.178
1790028163.975273845 push.services.mozilla.com 34.107.243.93
1790028164.325088349 firefox.settings.services.mozilla.com 140.248.169.91
1790028168.766933665 example.org 172.66.157.237,104.20.26.136
1790028168.766933996 example.org 172.66.157.237,104.20.26.136
1790028168.790920158 ipv4only.arpa 192.0.0.171,192.0.0.170
1790028168.793736736 detectportal.firefox.com 140.248.169.91
1790028168.793736812 detectportal.firefox.com 140.248.169.91
1790028170.264408865 ipv4only.arpa 192.0.0.170,192.0.0.171
1790028184.025839335 example.com 172.66.147.243,104.20.23.154
Warning: program compiled against libxml 215 using older 214
1790028163.023451839 140.248.169.91 443
1790028163.023955964 34.149.226.178 443
1790028163.998470727 34.107.243.93 443
1790028164.327997355 140.248.169.91 443
1790028164.680958054 140.248.169.91 443
1790028164.711340987 34.107.243.93 443
1790028164.903059085 34.107.243.93 443
1790028168.795090043 140.248.169.91 80
1790028170.273378067 140.248.169.91 80
1790028170.522573760 140.248.169.91 80

```

Figure 13: Full Browser Traffic Capture & Connection Extraction — Capturing complete browser traffic on eth0 and logging DNS answers alongside TCP connection requests.

```

1790028170.273378067 140.248.169.91 80
1790028170.522573760 140.248.169.91 80
1790028184.033492517 172.66.147.243 443
1790028184.036146160 172.66.147.243 443
1790028163.021495330 contile.services.mozilla.com 140.248.169.91
1790028163.021495365 content-signature-2.cdn.mozilla.net 34.149.226.178
1790028163.975273845 push.services.mozilla.com 34.107.243.93
1790028164.325088349 firefox.settings.services.mozilla.com 140.248.169.91
1790028168.766933665 example.org 172.66.157.237,104.20.26.136
1790028168.766933996 example.org 172.66.157.237,104.20.26.136
1790028168.790920158 ipv4only.arpa 192.0.0.171,192.0.0.170
1790028168.793736736 detectportal.firefox.com 140.248.169.91
1790028168.793736812 detectportal.firefox.com 140.248.169.91
1790028170.264408865 ipv4only.arpa 192.0.0.170,192.0.0.171
--
1790028163.023451839 140.248.169.91 443
1790028163.023955964 34.149.226.178 443
1790028163.998470727 34.107.243.93 443
1790028164.327997355 140.248.169.91 443
1790028164.680958054 140.248.169.91 443
1790028164.711340987 34.107.243.93 443
1790028164.903059085 34.107.243.93 443
1790028168.795090043 140.248.169.91 80
1790028170.273378067 140.248.169.91 80
1790028170.522573760 140.248.169.91 80

(kali@kali)-[~/SBT-DF203-Lab7]
$ echo "" == DNS Answers == | tee reports/dns_tcp_correlation.txt
cut -f3 reports/dns_A_answers_full.tsv | sort -u | tee -a reports/dns_tcp_correlation.txt
echo "" | tee -a reports/dns_tcp_correlation.txt
echo "" == TCP Destinations == | tee -a reports/dns_tcp_correlation.txt
cut -f2 reports/subsequent_tcp_destinations.tsv | sort -u | tee -a reports/dns_tcp_correlation.txt
echo "" | tee -a reports/dns_tcp_correlation.txt
echo "" == MATCHES (DNS answer IPs that appear in TCP destinations) == | tee -a reports/dns_tcp_correlation.txt
comm -12 <(cut -f3 reports/dns_A_answers_full.tsv | tr ' ' '\n' | sort -u) <(cut -f2 reports/subsequent_tcp_destinations.tsv | sort -u) | tee -a reports/dns_tcp_correlation.txt
cat reports/dns_tcp_correlation.txt
== DNS Answers ==
140.248.169.91
172.66.147.243,104.20.23.154
172.66.157.237,104.20.26.136

```

Figure 14: Automated DNS-to-TCP Connection Correlation Script — Executing comm -12 correlation matching DNS answer IPs against destination TCP addresses.

```
172.66.157.237,104.20.26.136
192.0.0.170,192.0.0.171
192.0.0.171,192.0.0.170
34.107.243.93
34.149.226.178

=== TCP Destinations ===
140.248.169.91
172.66.147.243
34.107.243.93
34.149.226.178

=== MATCHES (DNS answer IPs that appear in TCP destinations) ===
140.248.169.91
172.66.147.243
34.107.243.93
34.149.226.178

=== DNS Answers ===
140.248.169.91
172.66.147.243,104.20.23.154
172.66.157.237,104.20.26.136
192.0.0.170,192.0.0.171
192.0.0.171,192.0.0.170
34.107.243.93
34.149.226.178

=== TCP Destinations ===
140.248.169.91
172.66.147.243
34.107.243.93
34.149.226.178

=== MATCHES (DNS answer IPs that appear in TCP destinations) ===
140.248.169.91
172.66.147.243
34.107.243.93
34.149.226.178

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

Figure 15: *Final Correlation Matches Output* — Final verification output proving 4 DNS answer IPs directly matched subsequent TCP connections.

10. Conclusions

This lab successfully established a forensic baseline for Domain Name System operations. Controlled dig queries and live packet dissections proved that transaction IDs provide reliable request-to-response matching over UDP port 53. Browser traffic analysis revealed that single webpage requests spawn extensive cascades of auxiliary lookups. Crucially, the correlation between resolved IP addresses and subsequent TCP SYN destinations demonstrated how DNS telemetry provides invaluable corroborative evidence in network forensic investigations.

11. Appendix – Commands and Hashes

```
# Query Resolver
cat /etc/resolv.conf
dig example.com A
dig example.com AAAA
dig example.com MX
dig example.com NS

# Live DNS Capture
tshark -i eth0 -f 'port 53' -w evidence/fresh_dig_dns.pcapng

# Extract Queries and Responses
tshark -r working/fresh_dig_dns_working.pcapng -Y 'dns.flags.response == 0' -T fields ...
tshark -r working/fresh_dig_dns_working.pcapng -Y 'dns.flags.response == 1' -T fields ...

# DNS-to-TCP Connection Correlation
tshark -i eth0 -w evidence/browser_full.pcapng
comm -12 <(cut -f3 reports/dns_A_answers_full.tsv | tr ',' '\n' | sort -u) <(cut -f2
reports/subsequent_tcp_destinations.tsv | sort -u)

# SMTP Cross-Correlation
tshark -r ~/SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields ...
```

Evidence Integrity: Cryptographic Hash Digest Preservation:

- **Fresh DNS Capture Hash:** 1a1a09c8ca8f0e99ab25b397090c87f181e9d7e4bccdcf8d766867d2180454be
- **Browser DNS Capture Hash:** fbf9dbb7a619d1c8024971ea6178f82219f27875d57c019cfdb88f3c9991cec0

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 21/09/2026